Module 3
Business Economics

Answers

The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer 1(a)

Expenditure approach:

Y = C + I + G + (EX – IM)

Answer 1(b)(i)

Multiplier $\left[= \frac{Change\ \ in\ \ \ GDP}{Initial\ \ change\ \ in\ \ spending}\right] = \frac{\$50b}{\$10b} = 5$

MPC $\left[= 1 - \frac{1}{Multiplier}\right] = 1 - \frac{1}{5} = 0.8$

MPS [= 1 – MPC] = 1 – 0.8 = 0.2

Answer 1(b)(ii)

		(2)	(3)
	(1)	**Change in Consumption (in billion)**	**Change in Saving (in billion)**
	Change in Income (in billion)	**MPC (= 0.8)**	**MPS (= 0.2)**
Increase in investment of $10 billion	$10	$8	$2
Second round	$8	$6.4	$1.6
Third round	$6.4	$5.12	$1.28
All other rounds	$25.6	$20.48	$5.12
Total:	**$50**	$40	$10

Answer 1(b)(iii)

The larger the marginal propensity to consume (MPC), the larger the multiplier effect and the change in aggregate output/income will be for a given initial change in spending.

The larger the marginal propensity to save (MPS), the lower the multiplier effect and the change in aggregate output/income will be for a given initial change in spending.

Answer 1(b)(iv)

Factors that influence marginal propensity to consume include:

1. Interest rate: The higher the interest rate, the higher the incentive for individuals to save and the lower the marginal propensity to consume.

2. Inflation rate: The higher the inflation rate, the lower the money is worth in terms of purchasing power in the future, thereby prompting individuals to consume more of their income now.

3. Age (life cycle): Young people tend to consume more than old people because, for example, they are less concerned about their retirement planning. The higher the proportion of young people in the society, the higher the marginal propensity to consume will be.

4. Social security: The better the social security program, the lesser the need for people to save and the higher the marginal propensity to consume.

[Other valid answers are acceptable]

Answer 1(b)(v)

In an open economy with government tax, households would use some of the extra income in each subsequent round of consumption spending to purchase additional goods from abroad (imports) and pay additional taxes. This drains off some of the consumption spending (on domestic output) created by the increase in income. As a result, the multiplier effect will be smaller.

Answer 2(a)

FALSE. A decrease in the supply of RAM chips (supply curve shifts to the left) will not cause any shortage in RAM chips as long as prices of RAM chips are allowed to rise when supply decreases. The market for RAM chips will clear at a higher price when supply decreases.

Answer 2(b)(i)

$$E_D = \left(\frac{P_{\text{old}}}{Q_{\text{old}}}\right)\left(\frac{Q_{\text{new}} - Q_{\text{old}}}{P_{\text{new}} - P_{\text{old}}}\right)$$

$$-2 = \frac{\$100}{5{,}000} \times \frac{(Q_{\text{new}} - 5{,}000)}{-\$10} \Rightarrow Q_{\text{new}} = 6{,}000$$

Q_{new} = 6,000

It follows that the new quantity demanded is 6,000 packs.

Answer 2(b)(ii)

The original revenue was $100 × 5,000 = $500,000. The new revenue is $90 × 6,000 = $540,000. It follows that the new revenue is greater than the old revenue by $40,000.

Answer 2(b)(iii)

Since E_D = –1 suggests a uni-elastic demand (while E_D = –2 an elastic demand) over the sales range, the total revenue would remain unchanged.

Answer 2(c)(i)

Demand for poultry decreases.

This will lead to a fall in the equilibrium price and quantity.

As a price-taker, the decrease in market price would make the profit-maximizing farmer reduce its output.

Answer 2(c)(ii)

As farmers suffer losses, less-competitive farmers will leave the market in the long run.

This will decrease the market supply (lead to a fall in the market output) and thus a rise in the market price.

As market price rises, the output of individual farmers will rise as well.

Answer 3(a)

The confidence interval is a range of values around which the statistic calculated for a sample of observations (e.g. mean) is believed with a certain probability (e.g. 95%) to contain the true value of that statistic (i.e. the population value).

Answer 3(b)

The 95% confidence interval is given by

$$\left(\frac{580}{1500}\right) \pm (1.96)\sqrt{[(580/1500)(1- \quad 580/1500)]/1500}$$

0.3867 ± 0.0246

So, the 95% confidence interval is (0.3621, 0.4113).

Answer 3(c)

The researcher is 95% confident that the true proportion of households in the city which own a made-in-Japan flat screen TV is between 36.21% and 41.13%.

Answer 3(d)

Narrower.

Answer 4(a)

A parameter is a (numerical) measure used to describe a characteristic of a population.

A sample statistic is a (numerical) measure used to describe a characteristic of a sample.

Answer 4(b)

Population mean (measuring central tendency) and population standard deviation (measuring dispersion).

Answer 4(c)

Range of stock returns = 7% – (–7%) = 14%

Arithmetic average of stock returns

$= \left(\sum_{i=1}^{16} Xi\right) / 16 = 0$

Answer 4(d)

The median is the observation that divides a sample or population in half.

Median = (0 + 0) / 2 = 0

The mode is the observation that occurs the most often.

Mode = 0

Answer 4(e)

Since the set of value of the data in the positive and negative regions are exactly the mirror image of each other, the distribution's position can be fully determined by its mean (=0) and its shape being symmetric.

Answer 4(f)

$\text{SD} = \dfrac{\sum_{i=1}^{n}(Xi - \mu)}{n}$ where n is the population size and μ the population mean

$\text{SD} = [(7^2 + 6^2 + 5^2 + 4^2 + 3^2 + 2^2 + 1^2) \times 2 / 16]^{1/2}$
$= 4.18\%$

Answer 4(g)

The mean will decrease from 0% to –1% as each original value is subtracted by one.

The standard deviation, however, will remain unchanged as the squared deviation from the mean (i.e., $(x_i - \mu)^2$) remains unchanged.

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